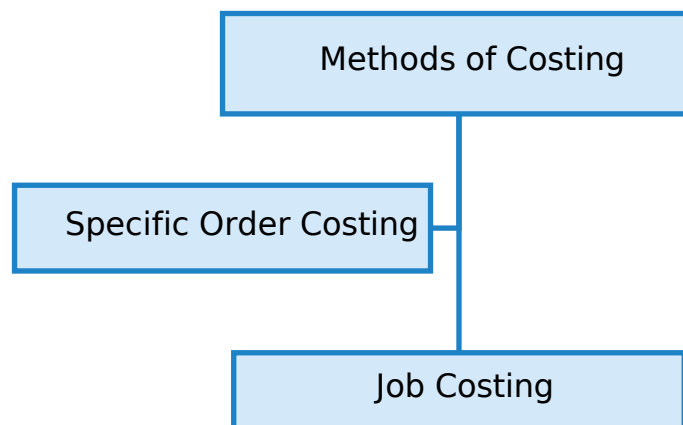


JOB COSTING



LEARNING OUTCOMES

- ♦ Describe Job Costing methods.
- ♦ Explain the accounting entries for cost elements under both the methods.
- ♦ Determining cost for a job.



1. JOB COSTING

1.1 Meaning of Job Costing

CIMA London defines Job Costing as **“the category of basic costing methods which is applicable where the work consists of separate contracts, jobs or batches, each of which is authorised by specific order or contract.”** According to this method, costs are collected and accumulated according to jobs, contracts, products or work orders. Each job or unit of production is treated as a separate entity for the purpose of costing. Job Costing is carried out for the purpose of ascertaining cost of each job and takes into account the cost of materials, employees and overhead etc. The job costing method is also applicable to industries in which production is carried out in batches. **Batch production basically is of the same character as the job order production, the difference being mainly one in the size of different orders.**

1.2 Principles of Job Costing

The job costing method may be regarded as the principal method of costing since the basic object and purpose of all costing is to:

- Analysis and ascertainment of cost of each unit of production
- Control and regulate cost
- Determine the profitability

The basic principles enunciated for the job costing method are valid essentially for all types of industry. For example, printing; furniture; hardware; ship-building; heavy machinery; interior decoration, repairs and other similar work.

1.3 Process of Job Costing

- Prepare a separate cost sheet for each job
- Disclose cost of materials issued for the job
- Employee costs incurred (on the basis of bill of material and time cards respectively)
- When job is completed, overhead charges are added for ascertaining total expenditure

Summary of costs	Estimated (₹)	Actual (₹)	For the job _____
Direct material cost			Units produced _____
Direct wages			Cost/unit _____
Production			Remarks _____
PRODUCTION COST			Prepared by: _____
Administration and			Checked by: _____
Selling &			
Distribution			
Overheads			
TOTAL COST			
PROFIT/LOSS			
SELLING PRICE			

3. COLLECTION OF COSTS FOR A JOB

3.1 Collection of Materials Cost

An essential requirement of job cost accounting is that **direct materials and their cost must be traced to and identified with specific job or work order**. This segregation of materials cost by jobs or work order is brought by the use of separate stores requisitions for each job or work order. Where a bill of material is prepared, it provides the basis for the preparation of these stores requisitions. But when the entire quantity of materials specified in the bill of materials is drawn in one lot or in installments, the bill itself could be made to serve as a substitute for the stores requisition.

After the materials have been issued and the stores requisitions have been priced, it is usual to enter the value of the stores requisition in a material abstract or analysis book. It serves to analyse and collect the cost of all direct materials according to job or work orders and departmental standing orders or expense code numbers. From the abstract book, **the summary of materials cost of each job is posted to individual job cost sheets or cards in the Work-in-Progress ledger**. The postings are usually made weekly or monthly. Similarly, at periodic intervals, from the material abstract books, summary cost of indirect material is posted to

different standing orders or expense code numbers in the Overhead Expenses ledger. If any special material has been purchased for a particular job, it is generally the practice to charge such special material direct to the job concerned without passing it through the Stores Ledger, as soon as it is purchased.

If any surplus material is left over in the case of any job, unless it can be immediately and economically used on some other job, the same is returned to the stores with a proper supporting document/stores Debit Note or Shop Credit, and the relevant job account is credited with the value of excess material returned to the stores. **If the surplus material is utilised on some other job, instead of being returned to the stores first, a material transfer note is prepared.** The transfer note would show the number of the transfer to job as well as transferee job (or jobs) so that, on that basis, the cost thereof can be adjusted in the Work-in-Progress Ledger.

3.2 Collection of Labour Cost

All direct labour cost must be analysed according to individual jobs or work orders. Similarly, different types of indirect labour cost also must be collected and accumulated under appropriate standing order or expenses code number. The analysis of labour according to jobs or work orders is, usually, made by means of job time cards or sheets. **All direct labour is booked against specific jobs in the job time cards or sheets.** All the idle time also is booked against appropriate standing order expense code number either in the job time card for each job or on a separate idle time card for each worker (where the job time card is issued job-wise). The time booked or recorded in the job time and idle time cards is valued at appropriate rates and entered in the labour abstract or analysis book. All direct employee cost is accumulated under relevant job or work order numbers, and **the total or the periodical total of each job or work order is then posted to the appropriate job cost card or sheet in Work-in-Progress ledger.** The postings are usually made at the end of each week or month.

The abstraction of idle time costs under suitable standing order or expenses code numbers is likewise done and the amounts are posted to the relevant departmental standing order or expense code number in the Overhead Expenses Ledger at periodical intervals. As regards other items of indirect labour cost these are collected from the payrolls books for the purpose of posting against standing order or expenses code numbers in the Overhead Expenses ledger.

3.3 Collection of Overheads

Manufacturing overheads are collected under suitable standing order numbers and selling and distribution overheads against cost accounts numbers. Total overhead expenses so collected are apportioned to service and production departments on some suitable basis. The expenses of service departments are finally transferred to production departments. The total overhead of production departments is then applied to products on some realistic basis, e.g. machine hour; labour hour; percentage of direct wages; percentage of direct materials; etc. It should be remembered that the use of different methods will lead to a different amount being computed for the works overhead charged to a job hence to different total cost. The problem of accurately absorbing, in each individual job or work order, the overhead cost of different cost centres or departments involved in the manufacture is difficult under the job costing method. It is because the cost or the expenses thereof cannot be traced to or identified with any particular job or work order. In such circumstances, the best that can be done is to apply a suitable overhead rate to each individual article manufactured or to each production order. This is essentially an *arbitrary* method.

3.4 Treatment of spoiled and defective work

Spoiled work is the quantity of production that has been totally rejected and cannot be rectified.

Defective work refers to production that is not as perfect as the saleable product but is capable of being rectified and brought to the required degree of perfection provided some additional expenditure is incurred. Normally, all the manufacturing operations are not fully successful; they result in turning out a certain amount of defective work. Nonetheless, over a period of time it is possible to work out a normal rate of defectives for each manufacturing process which would represent the number of defective articles which a process shall produce in spite of due care. Defects arise in the following circumstances:

Circumstances	Treatment
(1) Where a percentage of defective work is allowed in a particular batch as it cannot be avoided.	When a normal rate of defectives has already been established, if the actual number of defectives is within the normal limit or is near thereto the cost of rectification will be charged to the whole job and spread over the entire output of the batch. If, on the other hand, the number of defective units substantially exceeds the normal, the cost of rectification of the number which exceeds the normal will be written off as a loss in the Costing Profit and Loss Account.
(2) Where defect is due to bad workmanship.	In this case cost of rectification will be abnormal cost, <i>i.e.</i>, not a legitimate element of the cost. Therefore, the cost of rectification shall be written off as a loss, unless by an arrangement, it is to be recovered as a penalty from the workman concerned. It is possible, however that the management did provide for a certain proportion of defectives on account of bad workmanship as an unavoidable

	feature of production. If that be the case, the cost of rectifying to the extent provided for by the management will be treated as a normal cost and charged to the batch.
(3) Where defect is due to the Inspection Department wrongly accepting incoming material of poor quality.	In this case the cost of rectification will be charged to the department and will not be considered as cost of manufacture of the batch. Being an abnormal cost, it will be written off to the Costing Profit and Loss Account.



4. ACCOUNTING OF COSTS FOR A JOB

4.1 Entries in Control Accounts

1. For purchase of materials-
Stores Ledger Control A/c Dr.
 To Cost Ledger Control A/c*
2. For the value of direct materials issued to jobs-
Work-in-Process Control A/c Dr.
 To Stores Ledger Control A/c
3. For return of direct materials from jobs-
Stores Ledger Control A/c Dr.
 To Work-in-Process Control A/c
4. For return of materials to suppliers -
Cost Ledger Control A/c Dr.
 To Stores Ledger Control A/c
5. For indirect materials-
Factory Overhead Control A/c Dr.
 To Stores Ledger Control A/c
6. For wages paid-
Wages Control A/c Dr.
 To Cost Ledger Control A/c
7. For direct wages incurred on jobs-
Work-in-Process Control A/c Dr.
 To Wages Control A/c
8. For indirect wages -
Factory Overhead Control A/c Dr.
 To Wages Control A/c

- | | | |
|-----|--|-----|
| 9. | For any indirect expense paid-
Factory Overhead Control A/c
To Cost Ledger Control A/c | Dr. |
| 10. | For charging overhead to jobs-
Work-in-Process Control A/c
To Factory Overhead Control A/c | Dr. |
| 11. | For the total cost of jobs completed-
Cost of Sales A/c
To Work-in-Progress Control A/c | Dr. |
| 12. | The balance of Cost of Sales A/c is
transferred to Costing Profit and Loss
a/c; For such transfer -
Costing Profit and Loss A/c
To Cost of Sales A/c | Dr. |
| 13. | For the sales value of jobs completed -
Cost Ledger Control A/c
To Costing Profit and Loss A/c** | Dr. |

*General ledger adjustment account is another name of Cost Ledger Control Account.

****The balance of Costing Profit and Loss Account shall now represent profit or loss. The balance of Cost Ledger Control Account shall be carried forwarded. With the balance on all the accounts trial balance can be drawn.**

ILLUSTRATION 1

The manufacturing cost of a work order is ` 1,00,000; 8% of the production against that order spoiled and the rejection is estimated to have a realisable value of

2,000 only. The normal rate of spoilage is 2%. RECORD this in the costing journal.

SOLUTION

Actual loss due to spoilage = 8% of ` 1,00,000 = `8,000 and Normal loss = 2% of ` 1,00,000 = `2,000, therefore abnormal loss = `6,000.

The rejection has a realisable value of ` 2,000, which is to be apportioned between normal loss and abnormal loss in the ratio of 2 : 6.

The accounting entries necessary for recording the above facts would be:

		(`)	(`)
Material Control Account	Dr.	2,000	
Overhead Control Account	Dr.	1,500	
Costing Profit and Loss Control Account	Dr.	4,500	
To Work-in-Progress Control Account			8,000

In the case of defectives being inherent in the manufacturing process, the rectification cost may be charged to the specific jobs in which they have arisen. In case defectives cannot be identified with jobs, the cost of rectification may be treated as factory overheads. Abnormal defectives should be written off to the Costing Profit and Loss Account.

ILLUSTRATION 2

A shop floor supervisor of a small factory presented the following cost for Job No. 303, to determine the selling price.

	Per unit (`)
<i>Materials</i>	70
<i>Direct wages 18 hours @ ` 2.50 (Deptt. X 8 hours; Deptt. Y 6 hours; Deptt. Z 4 hours)</i>	45
<i>Chargeable expenses</i>	5
	120
<i>Add : 33-1/3 % for expenses cost</i>	40
	160

**Analysis of the Profit/Loss
Account (for the current
financial year)**

	(₹)		(₹)
Materials used	1,50,000	Sales less returns	2,50,000
Direct wages:			
Deptt. X	10,000		
Deptt. Y	12,000		
Deptt. Z	30,000		
	<u>8,000</u>		
Special stores items	4,000		
Overheads:			
Deptt. X	5,000		
Deptt. Y	9,000		
Deptt. Z	<u>2,000</u>		
	<u>16,000</u>		
Works cost	2,00,000		
Gross profit c/d	<u>50,000</u>		
	<u>2,50,000</u>		<u>2,50,000</u>
			<u>0</u>
Selling expenses	20,000	Gross profit b/d	50,000
Net profit	<u>30,000</u>		
	<u>50,000</u>		<u>50,000</u>

It is also noted that average hourly rates for the three Departments X, Y and Z are similar. You are required to:

- (i) PREPARE a job cost sheet.
- (ii) CALCULATE the entire revised cost using current financial year actual figures as basis.
- (iii) Add 20% to total cost to DETERMINE selling price.

SOLUTION

Job Cost Sheet

Customer Details ———

Job No. _____

Date of commencement ——

Date of completion _____

Particula	Amount
Direct materials	70
Direct wages:	
Deptt. X ` 2.50 × 8 hrs. = ` 20.00	
Deptt. Y ` 2.50 × 6 hrs. = ` 15.00	
Deptt. Z ` 2.50 × 4 hrs. = <u>` 10.00</u>	45
Chargeable expenses	<u>5</u>
Prime cost	120
Overheads:	
Deptt. X = $\frac{\text{`5,000}}{\text{`10,000}} \times 100 = 50\% \text{ of ` 20} = \text{` 10.00}$	
Deptt. Y = $\frac{\text{`9,000}}{\text{`12,000}} \times 100 = 75\% \text{ of ` 15} = \text{` 11.25}$	
Deptt. Z = $\frac{\text{`2,000}}{\text{`8,000}} \times 100 = 25\% \text{ of ` 10} = \text{` 2.50}$	<u>23.75</u>
Works cost	<u>143.75</u>
Selling expenses = $\frac{\text{`20,000}}{\text{`2,00,000}} \times 100 = 10\% \text{ of work cost}$	<u>14.38</u>
Total cost	158.13
Profit (20% of total cost)	<u>31.63</u>
Selling price	<u>189.76</u>

4.2 Advantages and Disadvantages of Job Costing

Some of the advantages and disadvantages of Job costing are summarised as below:

Advantages	Disadvantages
1. The details of Cost of material, labour and overhead for all job is available to control.	1. Job Costing is costly and laborious method.
2. Profitability of each job can be derived.	2. As lot of clerical process is involved the chances of error is more.
3. It facilitates production planning.	3. This method is not suitable in inflationary condition.
4. Budgetary control and Standard Costing can be applied in job costing.	4. Previous records of costs will be meaningless if there is any change in market condition.
5. Spoilage and defective can be identified and responsibilities can be fixed accordingly.	

4.3 Difference between Job Costing and Process Costing

The main points which distinguish job costing and process costing are as below:

Job Costing	Process Costing
(i) A Job is carried out or a product is produced by specific orders.	The process of producing the product has a continuous flow and the product produced is homogeneous.
(ii) Costs are determined for each job.	Costs are compiled on time basis i.e., for production of a given accounting period for each process or department.
(iii) Each job is separate and independent of other jobs.	Products lose their individual identity as they are manufactured in a continuous flow.

(iv) Each job or order has a number and costs are collected against the same job number.	The unit cost of process is an average cost for the period.
(v) Costs are computed when a job is completed. The cost of a job may be determined by adding all costs against the job.	Costs are calculated at the end of the cost period. The unit cost of a process may be computed by dividing the total cost for the period by the output of the process during that period.
(vi) As production is not continuous and each job may be different, so more managerial attention is required for effective control.	Process of production is usually standardized and is therefore, quite stable. Hence control here is comparatively easier.

SUMMARY

- ♦ **Job Costing:** The category of basic costing methods which is applicable where the work consists of separate contracts, jobs or batches, each of which is authorised by specific order or contract.

TEST YOUR KNOWLEDGE

Multiple Choice Questions (MCQs)

- In case product produced or jobs undertaken are of diverse nature, the system of costing to be used should be:*
 - Process costing*
 - Operating costing*
 - Job costing*
 - None of the above*
- The production planning department prepares a list of materials and stores required for the completion of a specific job order, this list is known as:*
 - Bin card*
 - Bill of material*

- (c) *Material requisition slip*
 - (d) *None of the above*
3. *Job costing is similar to that under Batch costing except with the difference that a:*
- (a) *Job becomes a cost unit.*
 - (b) *Batch becomes the cost unit instead of a job*
 - (c) *Process becomes a cost unit*
 - (d) *None of the above.*
4. *In job costing which of the following documents are used to record the issue of direct material to a job':*
- (a) *Goods received note*
 - (b) *Material requisition*
 - (c) *Purchase order*
 - (d) *Purchase requisition*
5. *The most suitable cost system where the products differ in type of materials and work performed is :*
- (a) *Job Costing*
 - (b) *Process Costing*
 - (c) *Operating Costing*
 - (d) *None of these.*
6. *Which of the following statements is true:*
- (a) *Job cost sheet may be used for estimating profit of jobs.*
 - (b) *Job costing cannot be used in conjunction with marginal costing.*
 - (c) *A production order is an order received from a customer for particular jobs.*
 - (d) *None of these.*

7. Which of the following statements is true:
- (a) Job cost sheet may be prepared for facilitating routing and scheduling of the job
 - (b) Job costing can be suitably used for concerns producing uniformly any specific product
 - (c) Job costing cannot be used in companies using standard costing
 - (d) Neither (a) nor (b) nor (c)

Theoretical Questions

1. DESCRIBE Job Costing giving example of industries where it is used.
2. DISTINGUISH between Job Costing & Process Costing.

Practical Problems

1. In a factory following the Job Costing Method, an abstract from the work-in- progress as on 30th September was prepared as under.

Job No.	Materials (`)	Direct hrs.	Labour (`)	Factory Overheads applied (`)
115	1325	400 hrs.	800	640
118	810	250 hrs.	500	400
120	765	300 hrs.	475	380
	2,900		1,775	1,420

Materials used in October were as follows:

Materials Requisition No.	Job No.	Cost (`)
54	118	300
55	118	425
56	118	515
57	120	665
58	121	910

59	124	720
		3,535

A summary for labour hours deployed during October is as under:

Job No.	Number of Hours	
	Shop A	Shop B
115	25	25
118	90	30
120	75	10
121	65	--
124	25	10
	275	75
<i>Indirect Labour: Waiting of material</i>	20	10
<i>Machine breakdown</i>	10	5
<i>Idle time</i>	5	6
<i>Overtime premium</i>	6	5
	316	101

A shop credit slip was issued in October, that material issued under Requisition No. 54 was returned back to stores as being not suitable. A material transfer note issued in October indicated that material issued under Requisition No. 55 for Job 118 was directed to Job 124.

The hourly rate in shop A per labour hour is ` 3 per hour while at shop B, it is ` 2 per hour. The factory overhead is applied at the same rate as in September. Job 115, 118 and 120 were completed in October.

You are asked to COMPUTE the factory cost of the completed jobs. It is the practice of the management to put a 10% on the factory cost to cover administration and selling overheads and invoice the job to the customer on a total cost plus 20% basis. DETERMINE the invoice price of these three jobs?

2. *Ares Plumbing and Fitting Ltd. (APFL) deals in plumbing materials and also provides plumbing services to its customers. On 12th August, 2022, APFL received a job order for a students' hostel to supply and fitting of plumbing materials. The work is to be done on the basis of specification provided by the hostel owner. Hostel will be inaugurated on 5th September, 2022 and the work is to be completed by 3rd September, 2022. Following are the details related with the job work:*

Direct Materials

APFL uses a weighted average method for the pricing of materials issues. Opening stock of materials as on 12th August 2022:

- 15mm GI Pipe, 12 units of (15 feet size) @ ` 600 each*
- 20mm GI Pipe, 10 units of (15 feet size) @ ` 660 each*
- Other fitting materials, 60 units @ ` 26 each*
- Stainless Steel Faucet, 6 units @ ` 204 each*
- Valve, 8 units @ ` 404*

each Purchases:

On 16th August 2022:

- 20mm GI Pipe, 30 units of (15 feet size) @ ` 610 each*
- 10 units of Valve @ ` 402*

each On 18th August 2022:

- Other fitting materials, 150 units @ ` 28 each*
- Stainless Steel Faucet, 15 units @ `*

209 each On 27th August 2022:

- 15mm GI Pipe, 35 units of (15 feet size) @ ` 628 each*
- 20mm GI Pipe, 20 units of (15 feet size) @ ` 660 each*
- Valve, 14 units @ ` 424*

each Issues for the hostel job:

On 12th August 2022:

- 20mm GI Pipe, 2 units of (15 feet size)
- Other fitting materials, 18

units On 17th August 2022:

- 15mm GI Pipe, 8 units of (15 feet size)
- Other fitting materials, 30

units On 28th August 2022:

- 20mm GI Pipe, 2 units of (15 feet size)
- 15mm GI Pipe, 10 units of (15 feet size)
- Other fitting materials, 34 units
- Valve, 6 units

On 30th August

2022:

- Other fitting materials, 60 units
- Stainless Steel Faucet, 15 units

Direct Labour:

Plumber: 180 hours @ ` 50 per hour (includes 12 hours overtime) Helper: 192 hours @ ` 35 per hour (includes 24 hours overtime) Overtimes are paid at 1.5 times of the normal wage rate.

Overheads:

Overheads are applied @ ` 13 per labour hour.

Pricing policy:

It is company's policy to price all orders based on achieving a profit margin of 25% on sales price.

You are required to

- Calculate the total cost of the job.
- Calculate the price to be charged from the customer

ANSWERS/ SOLUTIONS

Answers to the MCQs

1.	(c)	2.	(b)	3.	(a)	4.	(b)	5.	(a)	6.	(a)
7.	(d)										

Answers to the Theoretical Questions

1. Please refer paragraph 1
2. Please refer paragraph 4

Answers to the Practical Problems

1. **Factory Cost Statement of Completed Job.**

Month	Job No.	Materials	Direct labour	Factor y overheads (80% of direct labour cost)	Factory cost
	(`)	(`)	(`)	(`)	(`)
September	115	1,325	800	640	2765
October	115	--	125	100	225
Total		1,325	925	740	2,990
September	118	810	500	400	1,710
October	118	515	330	264	1,109
Total		1,325	830	664	2,819
September	120	765	475	380	1,620
October	120	665	245	196	1,106
Total		1,430	720	576	2,726

Invoice Price of Complete Job

Job No.	115 (`)	118 (`)	120 (`)
Factory cost	2,990.00	2,819.00	2,726.00
Administration and selling overheads @ 10% of factory cost	299.00	281.90	272.60
Total cost	3,289.00	3,100.90	2,998.60
Profit (20% of total cost)	657.80	620.18	599.72
Invoice Price	3,946.80	3,721.08	3,598.32

Assumption: - Indirect labour costs have been included in the factory overhead which has been recovered as 80% of the labour cost.

2. (a) Calculation of Total Cost for the Hostel Job:

Particulars	Amount (`)	Amount (`)
Direct Material Cost:		
- 15mm GI Pipe (Working Note- 1)	11,051.28	
- 20mm GI Pipe (Working Note- 2)	2,588.28	
- Other fitting materials (Working Note-3)	3,866.07	
- Stainless steel faucet 15 units × $\frac{6 \times `204 + 15 \times `209}{21 \text{ units}}$	3,113.57	
- Valve 6 units × $\frac{8 \times `404 + 10 \times `402 + 14 \times `424}{32 \text{ units}}$	2,472.75	23,091.95
Direct Labour:		
- Plumber [(180 hours × ` 50) + (12 hours × ` 25)]	9,300.00	
- Helper [(192 hours × ` 35) + (24 hours × ` 25)]	7,140.00	16,440.00

` 17.5)]		
----------	--	--

- Overheads [$\text{₹ } 13 \times (180 + 192) \text{ hours}$]		4,836.00
Total Cost		44,367.95

(b) Price to be charged for the job work:

	Amount (₹)
Total Cost incurred on the job	44,367.95
Add: 25% Profit on Job Price $\frac{44,367.95}{75\%} \times 25\%$	14,789.32
	59,157.27

Working Note:

1. Cost of 15mm GI Pipe

Date		Amount (₹)
17-08-2022	8 units $\times$ ₹ 600	4,800.00
28-08-2022	10 units $\times$ $\frac{4 \times ₹ 600 + 35 \times ₹ 628}{39 \text{ units}}$	6,251.28
		11,051.28

2. Cost of 20mm GI Pipe

Date		Amount (₹)
12-08-2022	2 units $\times$ ₹ 660	1,320.00
28-08-2022	$\frac{8 \times ₹ 660 + 30 \times ₹ 610 + 20 \times ₹ 660}{58 \text{ units}}$	1,268.28
		2,588.28

3. Cost of Other fitting materials

Date		Amount (₹)
12-08-2022	18 units $\times$ ₹ 26	468.00
17-08-	30 units $\times$ ₹ 26	780.00

2022		
------	--	--

28-08-2022	$34 \text{ units} \times \frac{12 \times 26 + 150 \times 28}{162 \text{ units}}$	946.96
30-08-2022	$60 \text{ units} \times \frac{12 \times 26 + 150 \times 28}{162 \text{ units}}$	1,671.11
		3,866.07